

Fundamentals of Solid State Physics

The Nearly Free Electron Model

Xing Sheng 盛兴

Department of Electronic Engineering
Tsinghua University

xingsheng@tsinghua.edu.cn

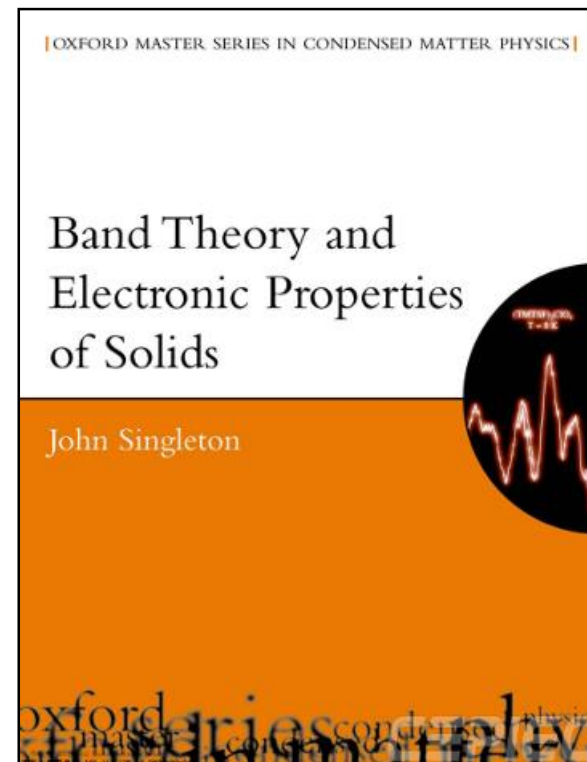
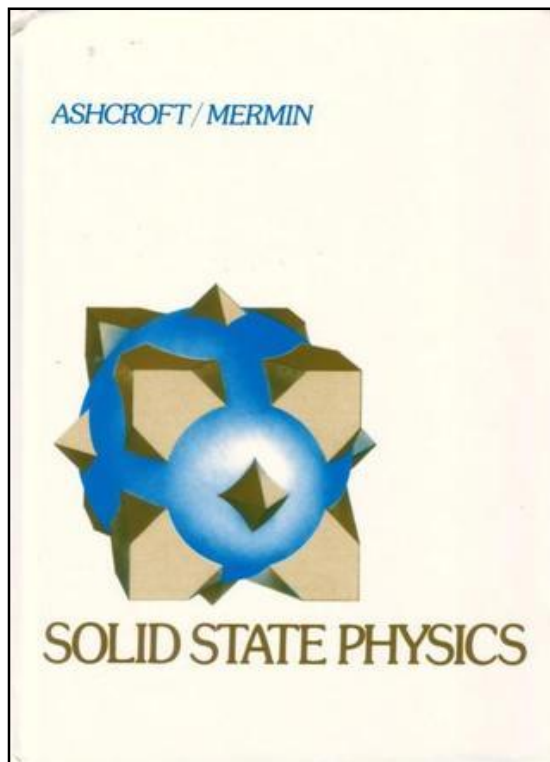


This Class

- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- **Electronic Properties (Week 4–12)**
 - Free electrons (the Drude and Sommerfeld models)
 - Electrons in a periodic potential (Bloch's Theorem)
 - The near-free electron model, the tight-binding model
 - Electronic band diagram, band gaps, effective mass
 - Metals, insulators, semiconductors
 - Devices: junctions, diodes, transistors
- Thermal Properties (Week 13)
- Optical Properties (Week 14)
- Magnetic Properties (Week 15)

Further Reading

- Ashcroft & Mermin, Chapter 9
- Singleton, Chapter 3



Real Electrons in Solids

Electrons are in *periodic* potentials

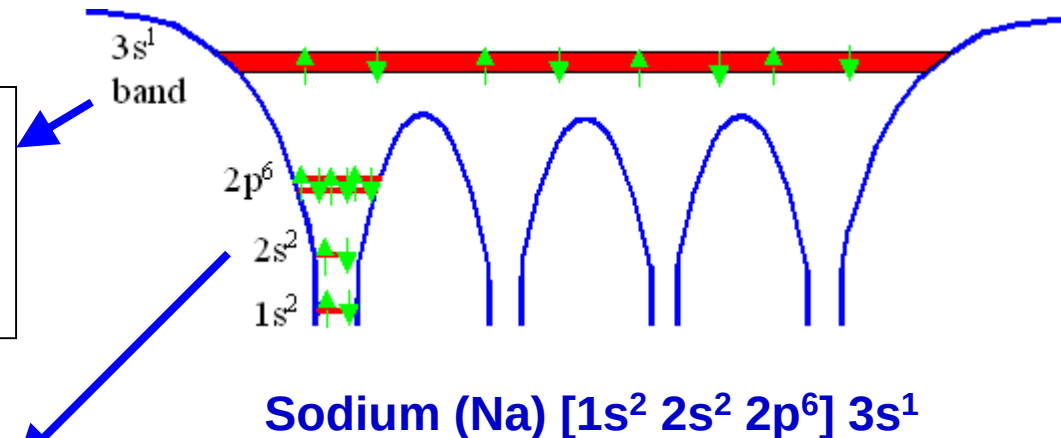


Bloch Wave

$$\psi(\mathbf{r}) = \exp(i\mathbf{k} \cdot \mathbf{r}) \cdot u_{\mathbf{k}}(\mathbf{r})$$

Nearly Free Electron Model
"近自由"近似

Tight Binding Model
"紧束缚"近似



An Electron in a Periodic Potential

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}) + V(\mathbf{r}) \cdot \psi(\mathbf{r}) = E \psi(\mathbf{r})$$

$$\psi(\mathbf{r}) = \sum_{\mathbf{k}} C_{\mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{r})$$

$$V(\mathbf{r}) = \sum_{\mathbf{G}} V_{\mathbf{G}} \exp(i\mathbf{G} \cdot \mathbf{r})$$

→
$$\left(\frac{\hbar^2 k^2}{2m} - E \right) C_{\mathbf{k}} + \sum_{\mathbf{G}} V_{\mathbf{G}} C_{\mathbf{k}-\mathbf{G}} = 0$$

The Central Equation

If we know $V(\mathbf{r})$ → $C_{\mathbf{k}}, C_{\mathbf{k} \pm \mathbf{G}}, C_{\mathbf{k} \pm 2\mathbf{G}} \dots$ $E_1, E_2, E_3 \dots$

The Central Equation

We only need to solve it in the first Brillouin zone

$$\begin{pmatrix} \ddots & & \dots & & \dots & & \ddots \\ \dots & \frac{\hbar^2}{2m}(k-g)^2 - E & & V_{-g} & & V_{-2g} & \dots \\ \dots & & & & & & \\ \dots & & V_g & & \frac{\hbar^2}{2m}k^2 - E & & \dots \\ \dots & & & & & & \\ \dots & & V_{2g} & & V_g & & \frac{\hbar^2}{2m}(k+g)^2 - E & \dots \\ \ddots & & \dots & & \dots & & \ddots \end{pmatrix} \begin{pmatrix} \vdots \\ C_{k-g} \\ C_k \\ C_{k+g} \\ \vdots \end{pmatrix} = 0$$

$V_{G=0}$ is a constant (the ground level energy)
here we set $V_{G=0} = 0$

$$g = \frac{2\pi}{a}$$

The Central Equation

We only need to solve it in the first Brillouin zone

$$\det \begin{pmatrix} \ddots & & & & \ddots \\ \dots & \frac{\hbar^2}{2m}(k-g)^2 - E & V_{-g} & V_{-2g} & \dots \\ \dots & V_g & \frac{\hbar^2}{2m}k^2 - E & V_{-g} & \dots \\ \dots & V_{2g} & V_g & \frac{\hbar^2}{2m}(k+g)^2 - E & \dots \\ \ddots & & & & \ddots \end{pmatrix} = 0$$

$$g = \frac{2\pi}{a}$$

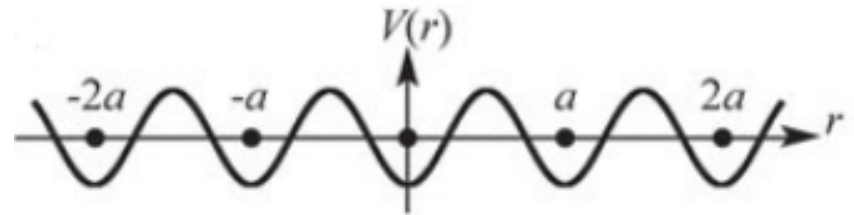
If we know $V(r)$ $\longrightarrow$

$$E_1(k), E_2(k), E_3(k), \dots$$

Nearly Free Electron Model

Consider a simple, weak periodic potential ($V \ll E$)

$$\begin{aligned} V &= -2V_1 \cos\left(\frac{2\pi}{a}x\right) \\ &= -2V_1 \cos(gx) \\ &= -V_1(e^{igx} + e^{-igx}) \end{aligned}$$



$$V(\mathbf{r}) = \sum_{\mathbf{G}} V_{\mathbf{G}} \exp(i\mathbf{G} \cdot \mathbf{r})$$



$$V_g = V_{-g} = -V_1$$

$$V_{2g} = V_{-2g} = V_{3g} = V_{-3g} = \dots = 0$$

The Central Equation

$$\det \begin{pmatrix} \ddots & \dots & \dots & \dots & \ddots \\ \dots & \frac{\hbar^2}{2m}(k-g)^2 - E & -V_1 & 0 & \dots \\ \dots & -V_1 & \frac{\hbar^2}{2m}k^2 - E & -V_1 & \dots \\ \dots & 0 & -V_1 & \frac{\hbar^2}{2m}(k+g)^2 - E & \dots \\ \ddots & \dots & \dots & \dots & \ddots \end{pmatrix} = 0$$



$$E_1, E_2, E_3 \dots$$

$$g = \frac{2\pi}{a}$$

The Central Equation

$$\det \begin{pmatrix} \ddots & \dots & \dots & \dots & \ddots \\ \dots & \frac{\hbar^2}{2m}(k-g)^2 - E & -V_1 & 0 & \dots \\ \dots & -V_1 & \frac{\hbar^2}{2m}k^2 - E & -V_1 & \dots \\ \dots & 0 & -V_1 & \frac{\hbar^2}{2m}(k+g)^2 - E & \dots \\ \ddots & \dots & \dots & \dots & \ddots \end{pmatrix} = 0$$



$$E_1, E_2, E_3 \dots$$

$$g = \frac{2\pi}{a}$$

The Central Equation

$$\begin{vmatrix} \frac{\hbar^2}{2m}(k-g)^2 - E & -V_1 \\ -V_1 & \frac{\hbar^2}{2m}k^2 - E \end{vmatrix} = 0$$

$$\rightarrow \left[\frac{\hbar^2}{2m}(k-g)^2 - E \right] \left[\frac{\hbar^2}{2m}k^2 - E \right] - V_1^2 = 0$$

$$\rightarrow \boxed{E_1(k), E_2(k)}$$

Energy Diagram

Free electron →

$$V_1 = 0$$

$$\left[\frac{\hbar^2}{2m} (k - g)^2 - E \right] \left[\frac{\hbar^2}{2m} k^2 - E \right] - V_1^2 = 0$$

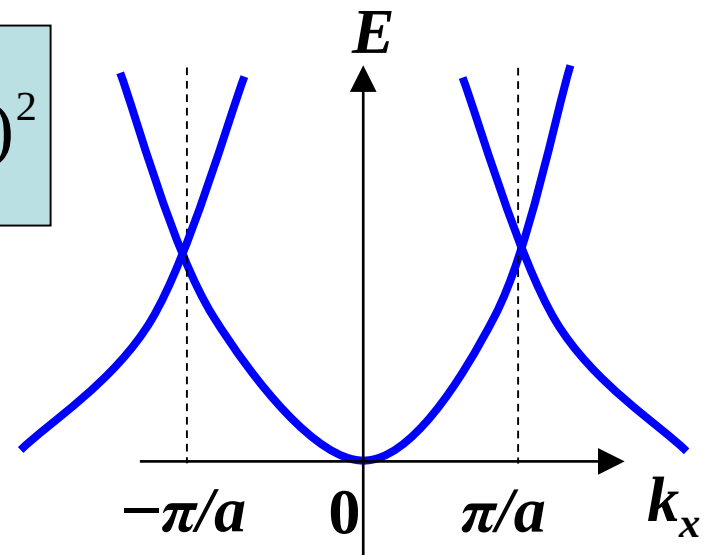
→

$$E = \frac{\hbar^2}{2m} k^2$$

or

$$E = \frac{\hbar^2}{2m} (k - g)^2$$

$$E \left(k = \frac{\pi}{a} \right) = \frac{\hbar^2}{2m} \left(\frac{\pi}{a} \right)^2$$



Energy Diagram

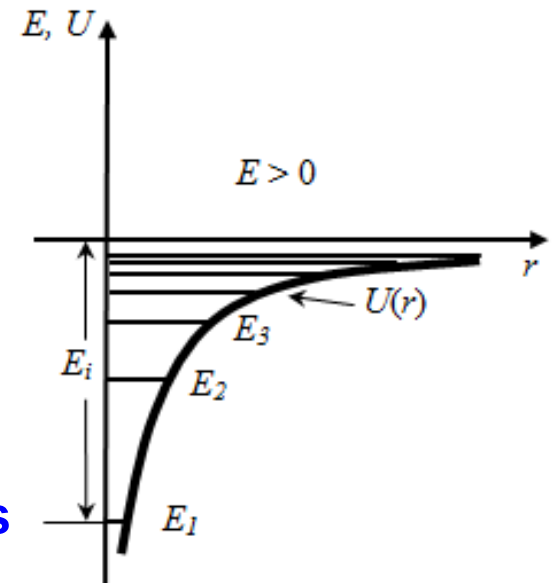
very strong potential $\rightarrow V_1 \gg 0$

$$\left[\frac{\hbar^2}{2m} (k - g)^2 - E \right] \left[\frac{\hbar^2}{2m} k^2 - E \right] - V_1^2 = 0$$

E is constant, independent of k

$$\begin{aligned} E_+ &= V_1 \\ E_- &= -V_1 \end{aligned}$$

discrete energy levels
(atomic orbitals)



Energy Diagram

Nearly Free electron

$$V_1 \neq 0$$

and

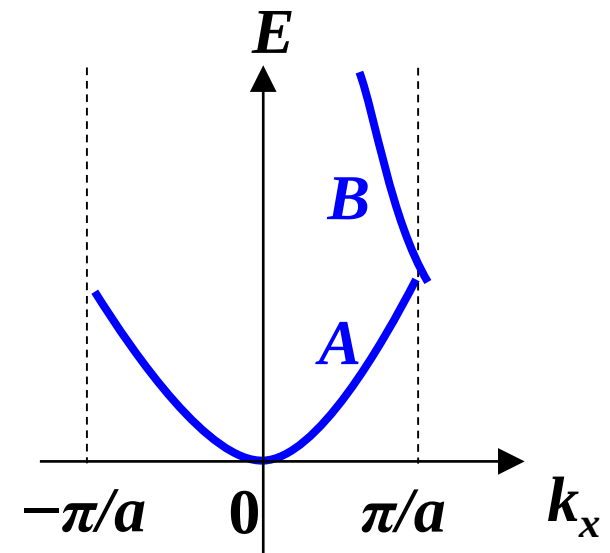
$$V_1 \ll \frac{\hbar^2}{2m} \left(\frac{\pi}{a} \right)^2$$

$$\left[\frac{\hbar^2}{2m} (k - g)^2 - E \right] \left[\frac{\hbar^2}{2m} k^2 - E \right] - V_1^2 = 0$$

$$E_{\pm} = \frac{(A + B) \pm \sqrt{(A - B)^2 + 4V_1^2}}{2}$$

$$A = \frac{\hbar^2}{2m} k^2$$

$$B = \frac{\hbar^2}{2m} (k - g)^2$$



A and B are solutions for free electrons

Energy Diagram

Nearly Free electron $\rightarrow$

$$V_1 \neq 0$$

and

$$V_1 \ll \frac{\hbar^2}{2m} \left(\frac{\pi}{a} \right)^2$$

$$\left[\frac{\hbar^2}{2m} (k - g)^2 - E \right] \left[\frac{\hbar^2}{2m} k^2 - E \right] - V_1^2 = 0$$

when $k \approx \pi/a$ $\rightarrow$

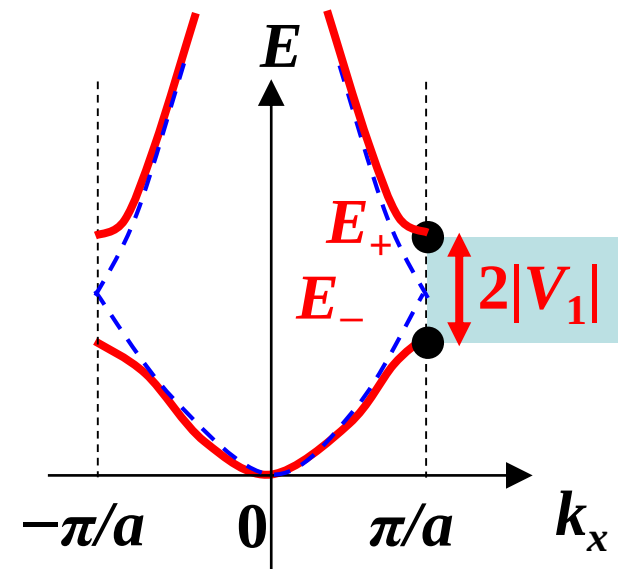
$$A \approx B$$

$$E_+ \approx B + |V_1|$$

higher than free electron

$$E_- \approx A - |V_1|$$

lower than free electron



Energy Diagram

Nearly Free electron $\rightarrow$

$$V_1 \neq 0$$

and

$$V_1 \ll \frac{\hbar^2}{2m} \left(\frac{\pi}{a} \right)^2$$

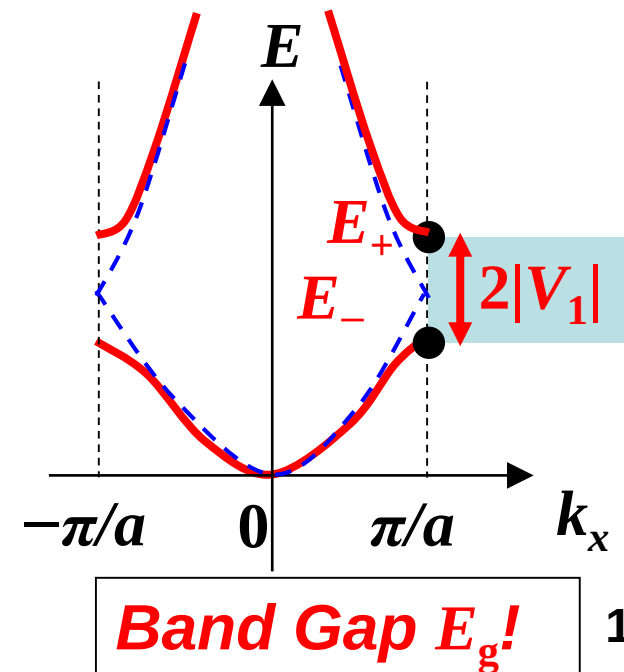
$$\left[\frac{\hbar^2}{2m} (k - g)^2 - E \right] \left[\frac{\hbar^2}{2m} k^2 - E \right] - V_1^2 = 0$$

when $k = \pi/a$ $\rightarrow$

$$A = B$$

$$E_+ \left(k = \frac{\pi}{a} \right) = \frac{\hbar^2}{2m} \left(\frac{\pi}{a} \right)^2 + |V_1|$$

$$E_- \left(k = \frac{\pi}{a} \right) = \frac{\hbar^2}{2m} \left(\frac{\pi}{a} \right)^2 - |V_1|$$

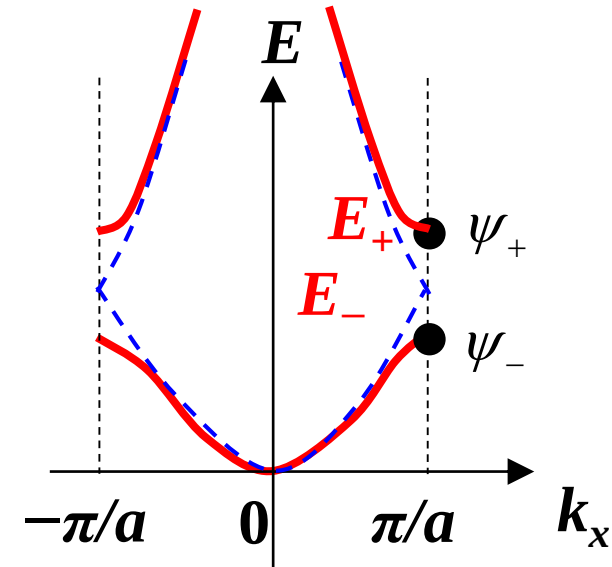


Energy Diagram

$$\begin{pmatrix} \frac{\hbar^2}{2m}(k-g)^2 - E & -V_1 \\ -V_1 & \frac{\hbar^2}{2m}k^2 - E \end{pmatrix} \begin{pmatrix} C_{k-g} \\ C_k \end{pmatrix} = 0$$

$$\psi(\mathbf{r}) = \sum_{\mathbf{k}} C_{\mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{r})$$

$$\psi = C_{k-g} \exp[i(k-g)x] + C_k \exp[ikx]$$



when $k = \pi/a$

$$E_- \left(k = \frac{\pi}{a} \right) = \frac{\hbar^2}{2m} \left(\frac{\pi}{a} \right)^2 - |V_1|$$



$$\psi_- \sim \cos(\pi x / a)$$

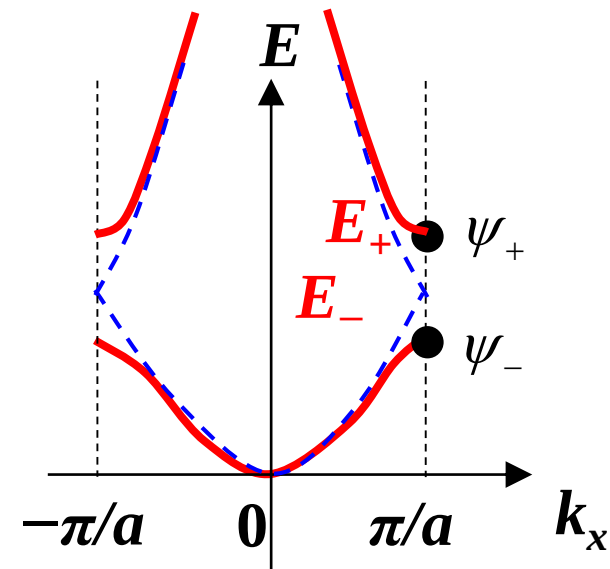
$$|\psi_-|^2 \sim \cos^2(\pi x / a)$$

Energy Diagram

$$\begin{pmatrix} \frac{\hbar^2}{2m}(k-g)^2 - E & -V_1 \\ -V_1 & \frac{\hbar^2}{2m}k^2 - E \end{pmatrix} \begin{pmatrix} C_{k-g} \\ C_k \end{pmatrix} = 0$$

$$\psi(\mathbf{r}) = \sum_{\mathbf{k}} C_{\mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{r})$$

$$\psi = C_{k-g} \exp[i(k-g)x] + C_k \exp[ikx]$$



when $k = \pi/a$

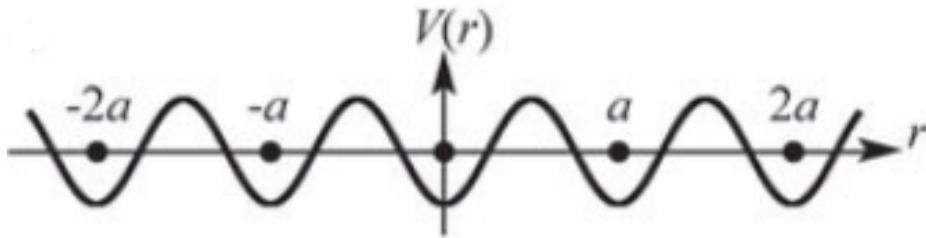
$$E_+ \left(k = \frac{\pi}{a} \right) = \frac{\hbar^2}{2m} \left(\frac{\pi}{a} \right)^2 + |V_1|$$



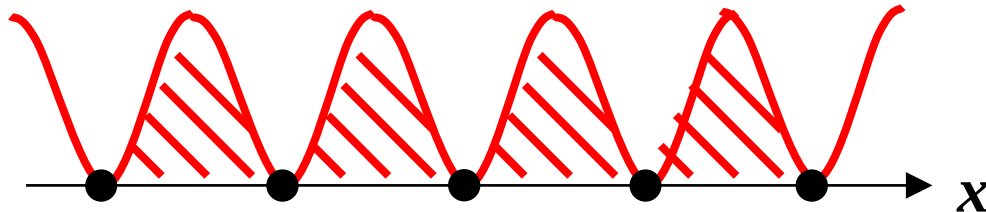
$$\psi_+ \sim i \sin(\pi x / a)$$

$$|\psi_+|^2 \sim \sin^2(\pi x / a)$$

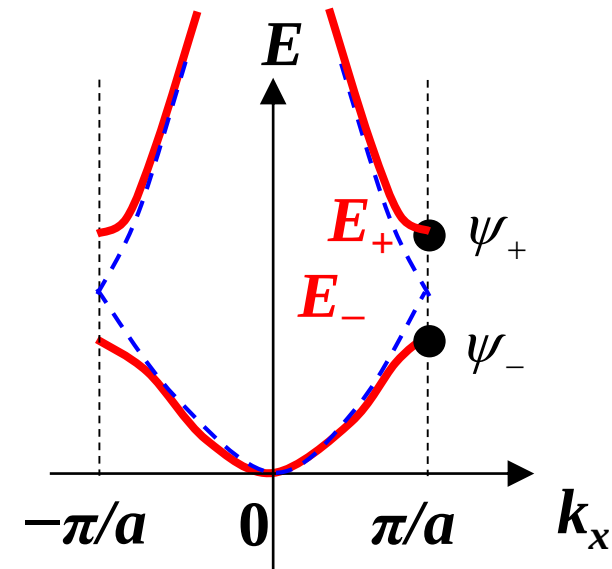
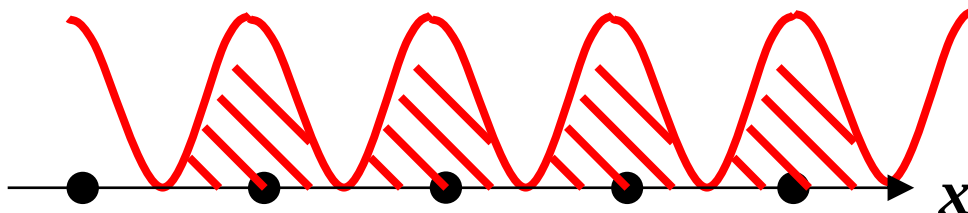
The Origin of Band Gap



$$|\psi(+)|^2 \sim \sin^2(\pi x / a)$$



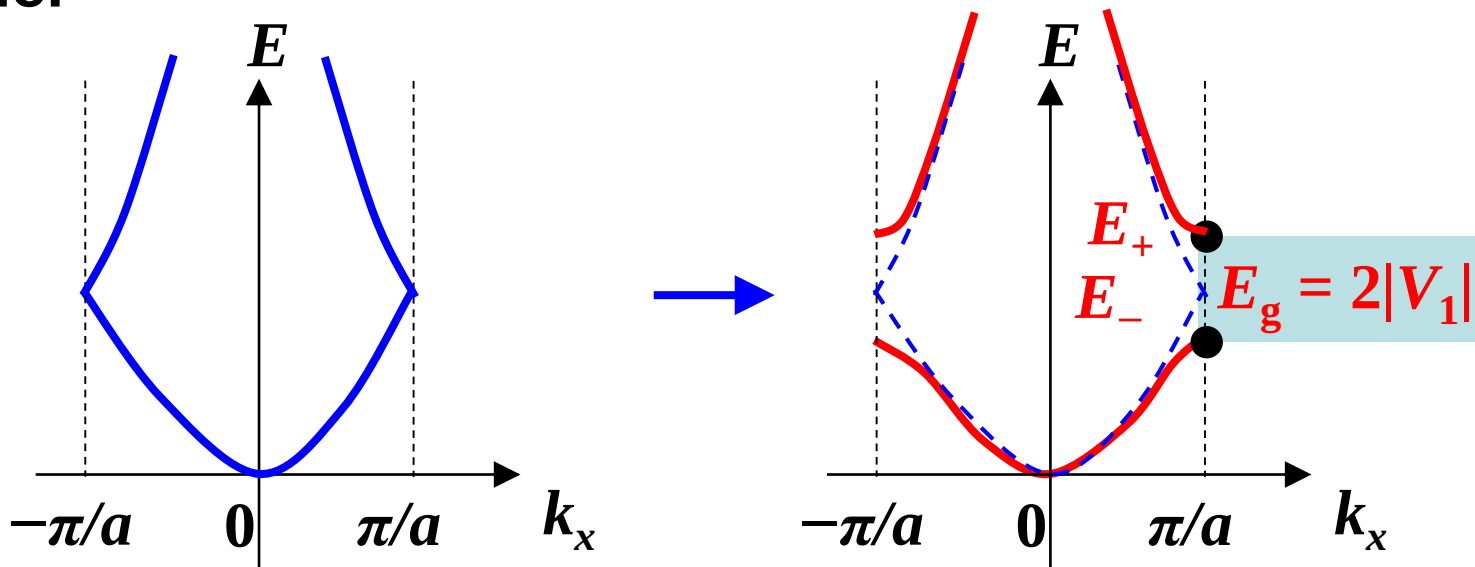
$$|\psi(-)|^2 \sim \cos^2(\pi x / a)$$



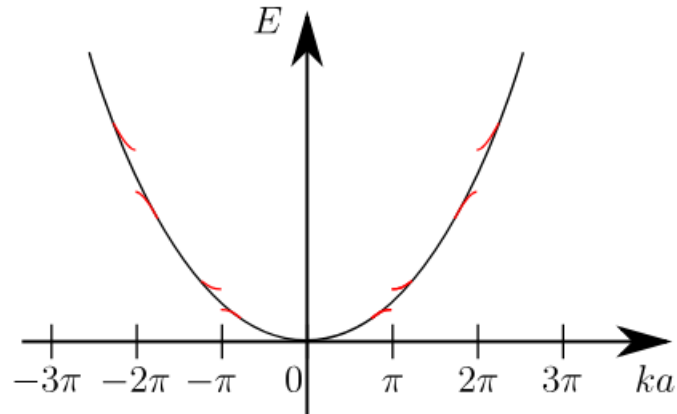
Two wavefunctions
have different
probability
distributions in the
lattice

The Origin of Band Gap

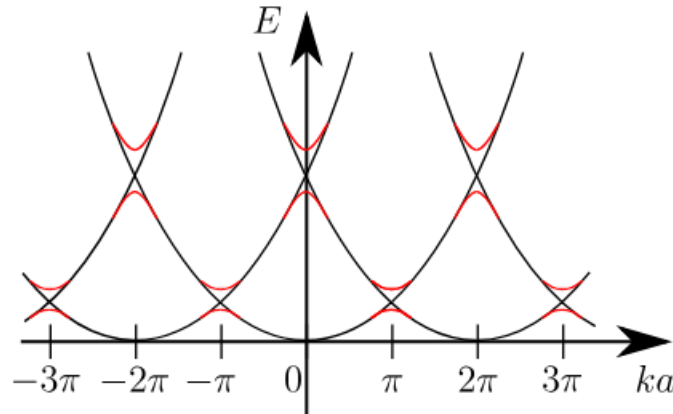
- Mathematical picture: The periodic potential perturbs the wavefunction of free electrons
- Physical picture: two electrons cannot occupy the same state (Pauli exclusion principle), the overlapped part has to be separated and repel each other



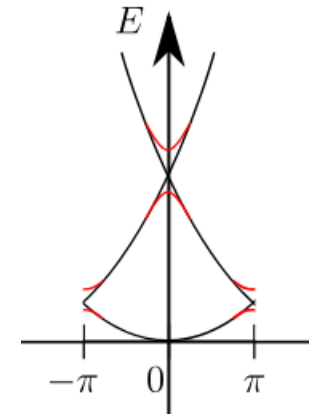
Band Structure / Diagram 能带图



**extended zone /
band structure**
扩展布里渊区

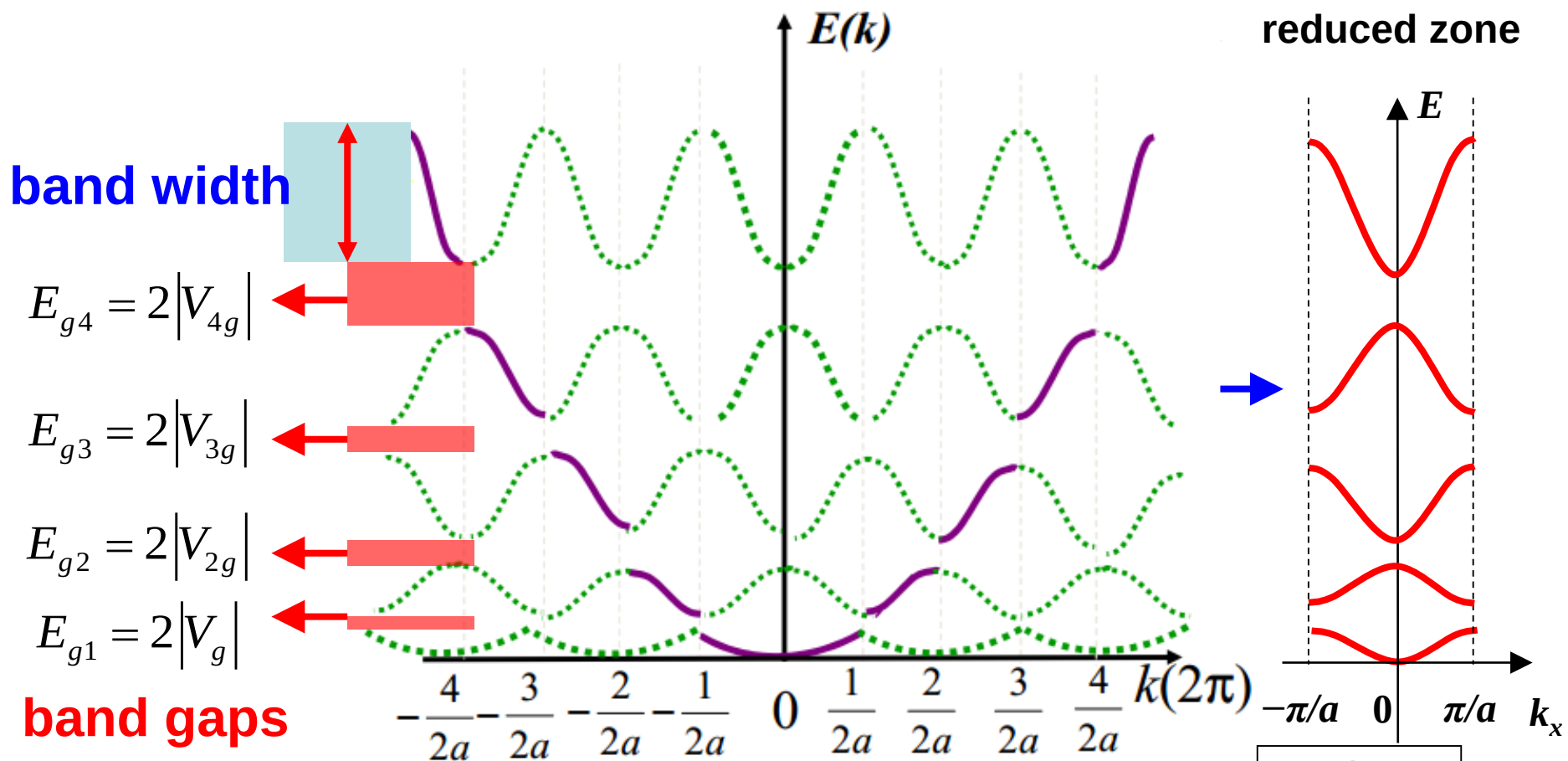


**repeated zone /
band structure**
周期性布里渊区



**reduced zone /
band structure**
简约布里渊区

Band Structure / Diagram



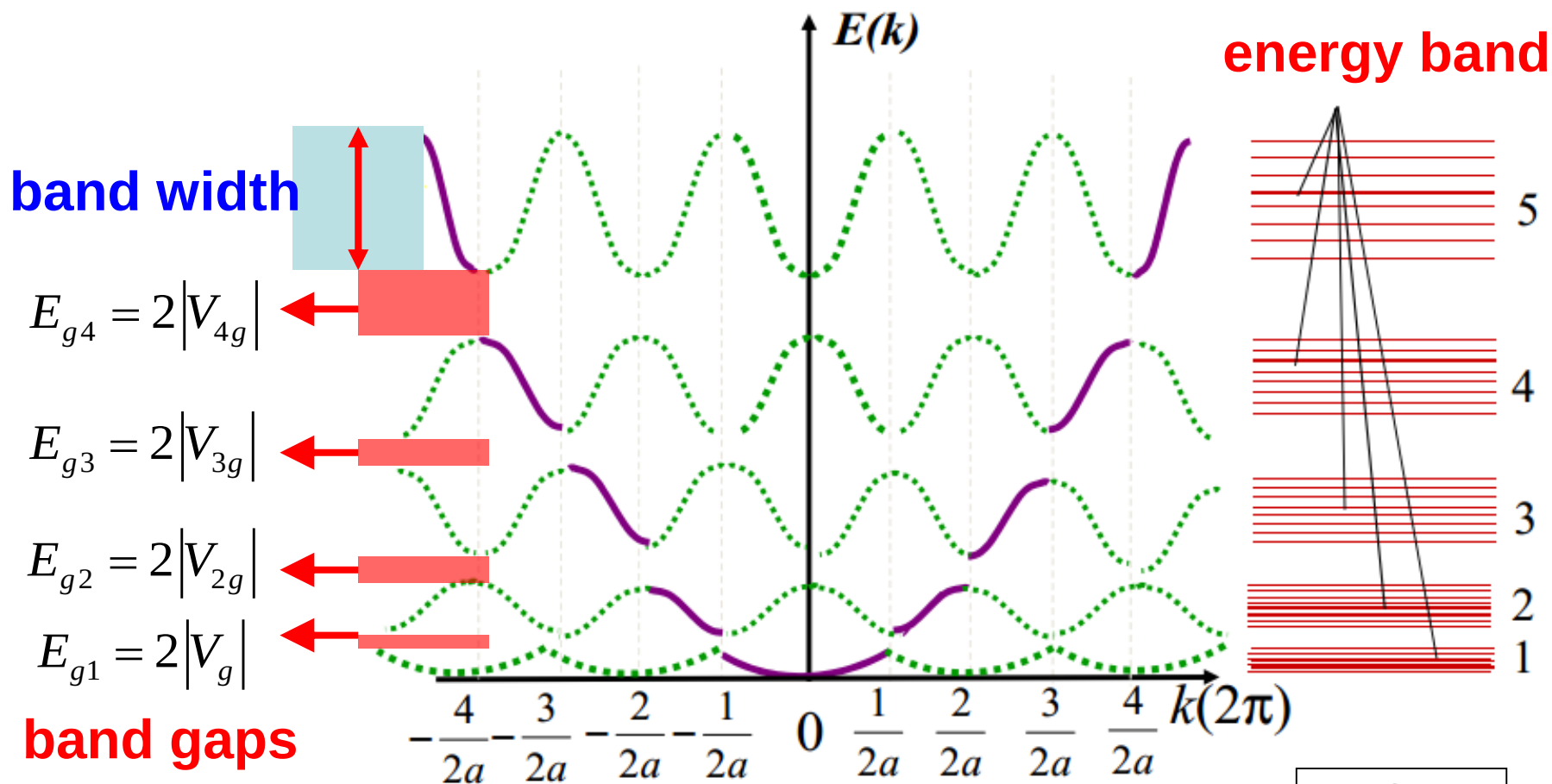
electron bands are quasi-continuous (准连续)

Q: how many k -states are there in each band?

$$k = \frac{2\pi}{a} \frac{n}{N}$$

$$n = 0, \pm 1, \pm 2, \dots \quad 22$$

Band Structure / Diagram



electron bands are quasi-continuous (准连续)

Q: how many k -states are there in each band?

$$k = \frac{2\pi}{a} \frac{n}{N}$$

$n = 0, \pm 1, \pm 2, \dots$ 23

Thank you for your attention